results for better-powered, non-pregistered placebo results for a specification where workers reported not seeing a score even when assigned to treatment, and when using lower-priority hospital visits as an outcome (columns 6 and 7 of Appendix Table A.6). Consistent with our pre-analysis plan, providing algorithm support specifically decreased hospitalizations for reasons more closely associated with maltreatment. We find an imprecise reduction in charges (cannot observe negotiated bill; column 5 of Appendix Table A.6).

We do not detect significant effects of providing the algorithmic tool on the incidence or number of re-referrals and removals (Appendix Table A.7). The absence of detectable impacts on removals should not be over-interpreted, as our study is underpowered to detect even large effects (e.g., power ≈ 0.2 to detect a 15 percent change). Moreover, the short-term impact of better-targeted investigations on removals is theoretically ambiguous. Our point estimates indicate a small possible reduction in the number of re-referrals (–5 percent), though the lack of statistical significance is more surprising for this outcome than for removals. This pattern suggests potential discrepancies between re-referral–based measures of maltreatment and hospital-based measures, reminiscent of the long-standing literature documenting differences between self-reported and substantiated CPS maltreatment (Negriff et al., 2017). Prior research has shown that the informational quality ("signal") of CPS referrals varies by reporter type (Nadon et al., 2023). In line with this evidence, when we restrict to re-referrals from law enforcement – a reporter type with a high substantiation rate and known to be relatively reliable – we find suggestive declines consistent with our hospital-based outcomes (–12.5 percent decrease in law enforcement re-referrals, $p = 0.20$). In contrast, re-referrals from less reliable reporters, such as education professionals, show no decline (+7.6 percent, $p = 0.47$). Given that CPS reporting is selective and intertwined with exposure to reporters, our findings suggest that re-referral counts may be an imperfect and noisy proxy for child welfare. Outcomes external to CPS, such as hospitalization, are therefore essential for corroborating the effectiveness of CPS interventions.

Exploring Spillovers and Learning. Importantly, we do not find evidence that tool access crowded out resources provided to children in the control group. Table A.8 tests for the presence of short-run, within-day spillovers (i.e., contamination) by leveraging the share of referrals that were treated during the day, excluding a child's own referral. If benefits had come at the expense of the control group, then, assuming fixed resources per day, one might expect to see: (1) worse outcomes for children in the control group as the percent treated in the day increases, and possibly also (2) weaker effects for treated children as the percent of children treated in the day increases. Though our confidence intervals are large, we find no evidence of

statistically significant spillover effects in terms of screen-ins, discussion time, or child harm. Overall, we interpret the results as suggestive, reassuring evidence that the algorithm benefited children in the treatment group without imposing costs on the control group.

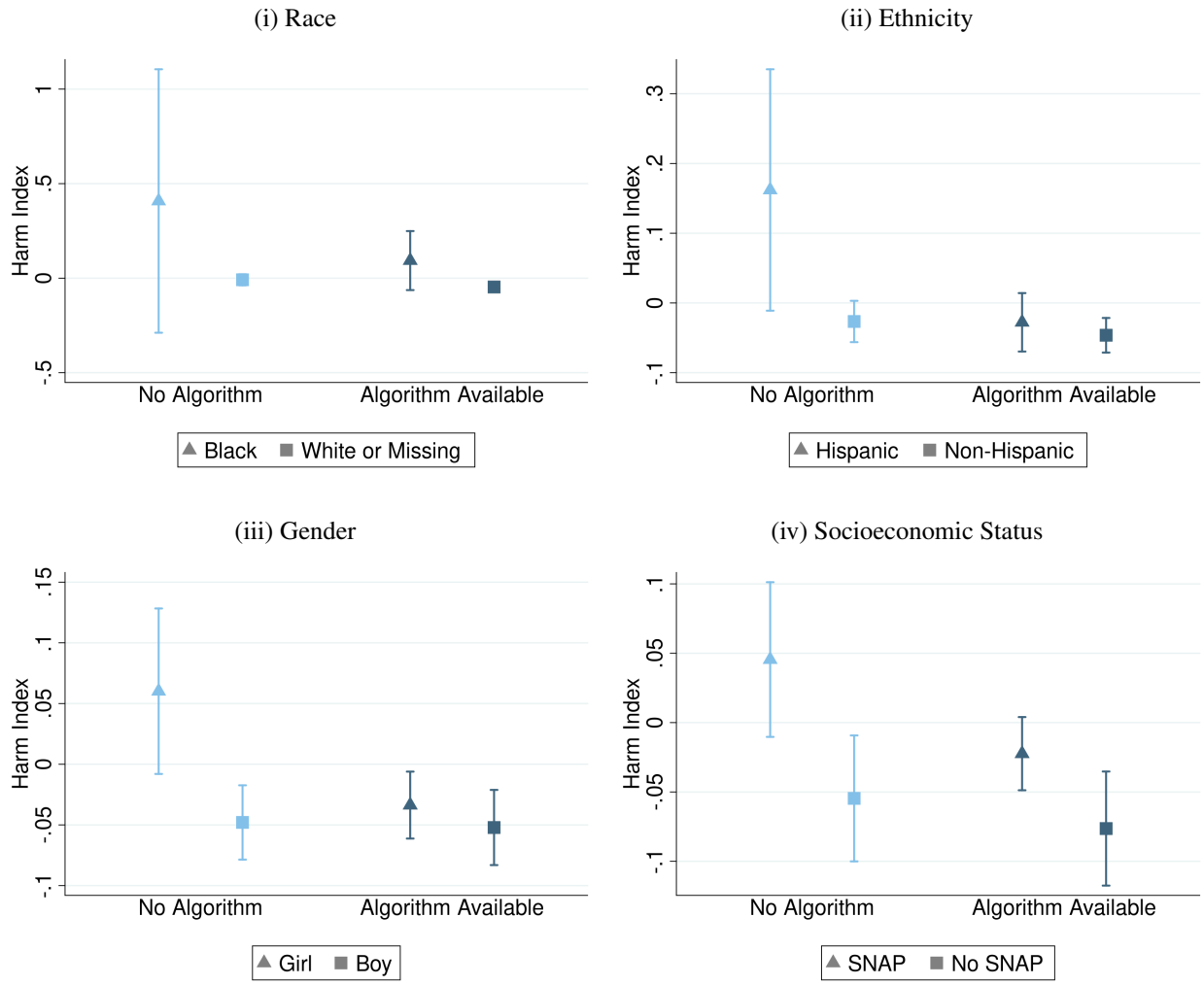
Our specification may underestimate the benefits of the algorithm if workers learned and changed their investigation tendencies from using the tool. In this scenario, we would expect indirect benefits for children in the control group and an attenuation of treatment effects over time. While our study is unable to assess learning that might have occurred after the end of the trial, we can observe differences in treatment effects for children referred at different points during the trial. Splitting the sample into four equally-sized groups by date of first referral in the trial, we find that harm is reduced by similar amounts for children referred in the first three quarters of the trial. There appears to be no impacts for children referred in the final quarter (Figure A.7), possibly owing to mechanical attenuation from a limited window to observe hospital outcomes for treated versus control during the trial (see footnote 11). The stability of estimates for the first three quarters of the trial are consistent with a lack of learning. However, these results should be interpreted with caution given the large confidence intervals by quarter and the fact that chronically-referred children were more likely to appear in earlier quarters of the trial. Although we cannot rule out learning within the trial, anecdotal evidence from interviews with workers suggest that workers did not consciously change their investigation strategy for control referrals. To the extent that there was unobserved learning by workers, our study estimates would likely be a conservative lower bound on the true impacts of providing the algorithm. Our discussions with supervisors at the end of the trial also suggest that workers did not become overly reliant on the tool, as supervisors correctly recalled that the tool only summarized history and not all dimensions of "future maltreatment risk."

4.2 Impact on Group-Level Disparities in Health and Wellbeing

In this section, we provide empirical evidence of how algorithm support affects disparities. A significant concern about algorithmic decision aids is their potential to perpetuate and widen inequality. Algorithms trained on past decisions have been repeatedly shown to embed historical bias (Obermeyer et al., 2019) especially when historical data was from automatic decision-making (Agan et al., 2022). They may also be less informative for members of disadvantaged or under-represented groups (Larrazabal et al., 2020; Blattner and Nelson, 2022). Humans have been documented to override algorithm recommendations in ways that reinforce existing biases (Albright, 2019, e.g.). On the other hand, algorithms may be *less* biased than status-

quo human decision-makers (c.f., Hoffman et al. 2018, Kleinberg et al. 2018, Arnold et al. 2024). Absence of rigorous empirical evidence on whether algorithmic tools indeed increase disparities has been a major hindrance to their adoption for social policy.

Figure 1: Algorithm Tool Reduced Harm the Most for Historically Disadvantaged Groups



Notes: These figures show the mean value of children's harm index by demographic category and tool availability, derived from regression estimates. SNAP (food stamp) receipt is a proxy for lower socioeconomic status. White children in Panel (i) have confidence intervals that are too small to be visible. 90% confidence intervals are reported and randomization controls are included throughout. Regression estimates with p-values are provided in Table A.9.

We provide some of the first field experiment evidence of the impact of algorithm tools on disparities in a social policy setting. Specifically, we report effects of algorithm availability on disparities in health by race, gender, ethnicity, and socioeconomic status (Figure 1 and Table A.9). When the tool was unavailable, as shown in light blue on the left side of each panel of Figure 1, children's harm index was elevated for Black children, Hispanic children, girls, and children from low-income families as measured by SNAP (food